

Work Experience

Simula Metropolitan Center for Digital Engineering (SimulaMet)

Oslo, Norway

PhD Student, Affiliated to Oslo Metropolitan University

May 2023 – Present

- Research on the potential applications of multimodal understanding across multiple domains.

The Alan Turing Institute

London, UK

PhD Enrichment Student

Oct 2024 – Jun 2025

- Student Representative for the 2024/25 cohort.

Thapathali Campus, Tribhuvan University

Kathmandu, Nepal

Adjunct Lecturer (Contract-based)

Oct 2022 – May 2023

- Lectures and labs for *A Practical Approach to Deep Learning*.
- Supervised MS projects related to deep learning.

North Star Developer's Village

Lalitpur, Nepal

Engineering Manager

Apr 2019 – May 2023

- ML face verification; pose-based exam cheating prevention; image/audio annotation; ML training platform.
- Semantic essay ranking; feedback bot; topic modelling/analysis; ML models exposed via APIs.

Leapfrog Technology

Kathmandu, Nepal

AI Intern

Oct 2018 – Mar 2019

- Predictive models for time-series weather and pollution forecasting (ARIMA and windowing-based).
- Model deployment and web-based interactive interface.

Department of Electronics & Computer Engineering (DOECE)

Lalitpur, Nepal

Student Research Assistant for UGC Research Grant Project

Jan 2018 – Sep 2018

- Assisted Prof. Dr. Nanda Bikram Adhikari on algorithm/tool research & implementation.
- Sensor data storage with GPS traces & data mining. Used RNN, RapidMiner, Hadoop, WebSocket, etc.

Education

Thapathali Campus, Tribhuvan University

Kathmandu, Nepal

Master's Degree in Informatics & Intelligent Systems Engineering

Oct 2020 – Oct 2022

Full Scholarship for Academic Merit, Top of Class

Relevant coursework: Natural Language Processing; Computer Vision; Applied Machine Learning; Deep Learning; Computational Intelligence; Information Theory; Information Visualization; Optimization.

Pulchowk Campus, Tribhuvan University

Lalitpur, Nepal

Bachelor's Degree in Computer Engineering

Oct 2015 – Oct 2019

Full Scholarship for Academic Merit

Relevant coursework: Artificial Intelligence; Probability & Statistics; Data Mining; Big Data; Theory of Computation.

Technical Capabilities

Programming: Python, Django, Android (Java/Kotlin), SQL, C/C++.

Libraries: PyTorch, TensorFlow/Keras, Pandas, NumPy, scikit-learn, SciPy, Matplotlib.

Languages: English (Fluent), Nepali (Native), Hindi (Advanced).

Academic Projects

MS Thesis: AI-Based Soccer Game Summarization: From Video Highlights to Dynamic Text Summaries.

Best thesis in the department (by scores). In collaboration with the Department of Holistic Systems, **SimulaMet**.

MS Project: Deep Learning-based Pose Estimation for Dystonia Score Prediction. In collaboration with **NAAMII**.

Bachelor's Thesis: Assessment of Spatiotemporal Weather and Pollution Proxies with RNN and ARIMA.

Best Project Award by Department.

Bachelor's Project: Human Facial Landmarks Detection using Multi-Layer Perceptron.

Selected Publications

- 2022:** Soccer Game Summarization using Audio Commentary, Metadata, and Captions. NarSUM, ACM Multimedia.
- 2022:** Assisting Soccer Game Summarization via Audio Intensity Analysis of Game Highlights. IOE Graduate Conference.
- 2020:** Near Real-Time Mobile Profiling and Modeling of Fine-Scale Environmental Proxies Along Major Road Lines of Nepal. Springer, Cham.
- 2019:** Sentence Ranking and Answer Pinpointing in Online Discussion Forums Utilizing User-generated Metrics and Highlights. NaSCoIT.
- 2018:** Real Time-Based Smart Traffic Light System with its Simulation using 8051 Microcontroller. KEC Conference.

Awards & Certifications

- NAAMII Winter School:** Full scholarship for participation. Topics: Deep Learning; Computer Vision; Statistics & Algorithms; Data Science; Optimization.
- Hult Prize TU Winner (2017):** Winning team "AEGIS". Presented a sustainable idea to empower refugees in Nepal using an e-commerce platform; presented at Hult Business School, Dubai.
- Mentor:** Global Venture Bootcamp, Asian Development Bank, Philippines — enhancing social ventures with emerging technologies.
- Participant:** IEEE AI & Big Data Workshop 2018; led multiple IEEE Student Chapter activities on campus.
- Participant:** UNICEF 1000 Golden Hours Code Camp (2017) — Best Prototype Award for mobile app idea.

References

- Prof. Pål Halvorsen:** Chief Research Scientist, SimulaMet; Professor, OsloMet; Adjunct Professor, UiO. paalh@simula.uio.no / paalh@ifi.uio.no; +47 97080007.
- Dr. Cise Midoglu:** Postdoctoral Fellow, Department of Holistic Systems (HOST), SimulaMet. cise@simula.no; +47 92287195.
- Asst. Prof. Dinesh B. Kshetri:** MSc Program Coordinator, IOE, Tribhuvan University. dinesh@ioe.edu.np; +977 9818951364.